

Chapter 3

Artificial Intelligence and Future of Systems Engineering



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Abstract Systems Engineering (SE) is in the midst of a digital transformation driven by advanced modeling tools, data integration, and resulting “digital twins.” Like many other domains, the engineering disciplines will see transformational advances in the use of artificial intelligence (AI) and machine learning (ML) to automate many routine engineering tasks. At the same time, applying AI, ML, and automation to complex and critical systems needs holistic, system-oriented approaches. This will encourage new systems engineering methods, processes, and tools. It is imperative that the SE community deeply understand emerging AI and ML technologies and applications, incorporate them into methods and tools, and ensure that appropriate SE approaches are used to make AI systems ethical, reliable, safe, and secure. This chapter presents a road mapping activity undertaken by the Systems Engineering Research Center (SERC). The goal is to broadly identify opportunities and risks that might appear as this evolution proceeds as well as potentially provide information that guides further research in both SE and AI/ML.

3.1 Introduction

In 2019, the Research Council of the Systems Engineering Research Center (SERC), a U.S. Defense Department sponsored University Affiliated Research Center (UARC), developed a roadmap to structure and guide research in artificial intelligence (AI) and autonomy. This roadmap was updated in 2020. This chapter presents the current roadmap as well as key aspects of the underlying Digital Engineering transformation that will enable both transformation of SE practices using AI for SE and drive the need for new systems engineering practices that support a new wave of automated, adaptive, and learning systems, termed SE for AI. The “AI4SE” and “SE4AI” labels have become metaphors for an upcoming rapid evolutionary phase in the SE Community.

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AI4SE applies AI and ML techniques to improve human-driven engineering practices. This goal of “augmented intelligence” includes outcomes such as achieving scale in model construction and efficiency in design space exploration. SE4AI applies SE methods to learning-based systems’ design and operation, with outcomes such as improved safety, security, ethics, etc.

SE is in the midst of a digital transformation driven by advanced modeling tools, data integration, and the resulting “digital twins” that maintain virtual copies of portions of real-world systems across lifecycles of system use. This transformation is changing what used to be primarily document-based system descriptions (concept of operations, requirements, architectures, etc.) into digital data and descriptive models that link data from different disciplines together. This central dataset, known as an “authoritative source of truth,” will over time integrate all aspects of engineering design, use, and maintenance of systems into a linked set of information. This digital engineering transformation will be followed by transformational advances in the discipline of systems engineering using AI and ML technology for automation of many engineering tasks, designed to augment human intelligence.

At the same time, the application of AI, ML, and autonomy to many of today’s complex and critical systems drives the need for new SE methods, processes, and tools. Today, applications of these technologies represent serious challenges to the SE community. A primary goal of SE is to ensure that the behavior and performance of complex engineered systems meet the expected outcomes driven by user needs, and that the configuration of the system is managed across its lifetime. Advances in AI and ML application mean that future system components may learn and adapt more rapidly, and that behavior and performance may be non-deterministic with less predictable but manageable outcomes. This change may introduce new failure modes not previously experienced in the engineering community. The inability to explicitly validate system behaviors or the time it takes to do that will impact trust in these systems and will change the way the SE community traditionally addresses system validation. The uncertainty present in multiple AI/ML components that interact will defy traditional decomposition methods used by the SE community, requiring new synthesis methods. Finally, as systems develop means for co-learning between human users and machines, traditional models that separate human behaviors from the machine will need to be revisited.

At an early 2019 Future of Systems Engineering (FuSE) workshop hosted by the International Council on Systems Engineering (INCOSE), the terms AI for SE and SE for AI were first used to describe this dual transformation (McDermott et al., 2020). The “AI4SE” and “SE4AI” labels have quickly become metaphors for an upcoming rapid evolutionary phase in the SE Community. AI4SE may be defined as the application of augmented intelligence and machine learning techniques to support the practice of systems engineering. Goals in such applications include achieving scale in model construction and confidence in design space exploration. SE4AI may be defined as the application of systems engineering methods to the design and operation of learning-based systems. Key research application areas include the development of principles for learning-based systems design, models of life cycle evolution, and model curation methods.

3.2 SERC AI4SE and SE4AI Roadmap

In order to better understand and focus on this evolution, the Research Council of the SERC developed a roadmap to structure and guide research in artificial intelligence (AI) and autonomy. This roadmap was described in McDermott et al. (2020) and presented in a number of forums including both systems disciplines (McDermott, 2019, 2020a) and AI disciplines (McDermott, 2020b). A dedicated “SE4AI/AI4SE” workshop sponsored by the SERC and the U.S. Army further refined the roadmap. An initial version was presented at the Fall 2020 Association for the Advancement of Artificial Intelligence (AAAI) conference, and the current version will be published in 2021 in an INCOSE AI primer for systems engineers. This roadmap is being published with a goal to link the discipline of systems engineering to various trends in artificial intelligence and its application to automation in systems. This linkage is provided as a means to discuss the possible evolution of AI/ML technology, autonomy, and the SE discipline over time. Figure 3.1 depicts the current notional roadmap.

The envisioned long-term outcome is “Human–Machine Co-learning.” This outcome captures a future where both humans and machines will adapt their behavior over time by learning from each other or alongside each other. For the SE community, this is a new context and lifecycle model that is not envisioned and supported by most of the current-day systems engineering practices. This new context implies a fairly significant transformation of SE methods, tools, and practices and is underway that will change both SE and AI methods, processes, and tools over time.

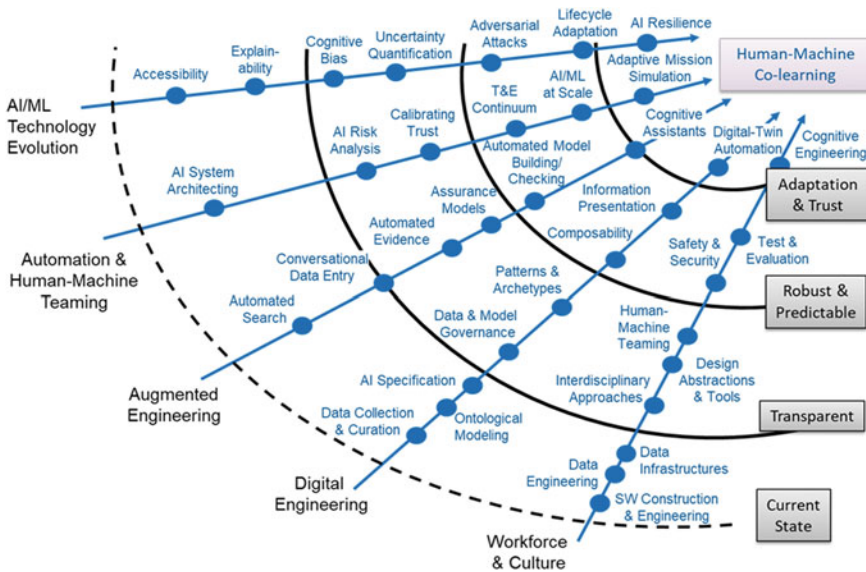


Fig. 3.1 The SERC AI4SE and SE4AI Roadmap

To achieve this end state, one might consider there is a need for both the AI and SE disciplines to pass through a set of “waves” or eras. The first of these includes sets of technologies and approaches that make the decisions produced by AI systems more transparent to the human developers and users. This broadly relates to the evolution of trust in such systems. Today, much of the activity in the “transparency” wave is termed “Explainable AI,” but it also includes more transparency and understanding of the methods and tools used to develop AI applications, the underlying data, and the human–machine interfaces that lead to effective decision-making in the type of complex systems SE deals with routinely.

The “robust and predictable” wave is to produce systems that learn and may be non-deterministic, but that is also appropriately robust, predictable, and trustworthy in the type of critical and complex uses common to the application of SE practices today. This wave particularly includes both human and machine behaviors in joint decision environments, highly reliant on good human-system design and presentation of decision information. It also includes the adaptation of test and evaluation processes to co-learning environments.

The third wave involves systems that actually adapt and learn dynamically from their environments. In this wave, machine-to-machine and human-to-machine (and maybe machine-to-human) trust will be critical. Trust implies a dependence between the human and machine, which must emerge from human–machine interaction. Trust normally requires the human to understand and validate the performance of the system against a set of criteria in a known context. In this third wave, systems will be expected to learn to modify or create new behaviors as the context changes and this may happen fairly rapidly. Methods that revalidate system performance extremely rapidly or “on the fly” are not part of the current SE practice set and must be developed along with these types of learning systems.

The vectors of this notional roadmap span five categories. The first of these vectors recognizes that the technological implementation of AI systems will evolve and need to evolve in directions relevant to SE. Most of these directions can be related to transparency and trust in the technology. The second vector recognizes that the purpose of AI in systems is generally to provide for automation of human tasks and decisions, and this will change how we design and test systems. The third vector recognizes that AI technologies will gradually be used more and more to augment the work of engineering. The fourth vector recognizes that the current digital engineering transformation will be an enabler for that. A short description of the first four categories is included in Tables 3.1, 3.2, 3.3, 3.4. The fifth vector recognizes a transformation will need to be accomplished in the SE workforce, with significantly more integration of software and human behavioral sciences at the forefront.

3.3 Digital Engineering

We start with digital engineering as it is the enabler for the first three roadmap vectors. Table 3.1 provides a consolidated summary of the research and development

Table 3.1 Digital engineering as the enabler for AI4SE and SE4AI

Research area	Definition	Use in systems engineering
Digital engineering		
Data collection and curation	Specific activities to build infrastructure and collect and manage data needed for engineering and programmatic activities in system development and support	Provides connectivity and reuse of data across disciplines and system lifecycles
Ontological modeling	Knowledge representation of engineering and programmatic data providing interoperability through standard and domain-specific ontologies	Improves model-based systems engineering; enables AI-based reasoning
AI specification (Seshia et al., 2018)	System-level and formal specifications for AI behaviors supporting verification activities	Allows specification of requirements for AI-related functions in a mathematically rigorous form
Data and model governance	Lifecycle management, control, preservation, and enhancement of models and associated data to ensure value for current and future use, as well as repurposing beyond initial purpose and context	Configuration management, quality management, pedigree, and reuse of digital data and models
Patterns and archetypes	Widely used modeling constructs that separate design from implementation, supporting better reuse and composition	Development of standard model templates and patterns for composition
Composability	Rapid development and integration of design using higher level abstracted components and patterns, across multiple disciplines	Development of domain-specific computer languages and low-code software tools
Information presentation	Visualization approaches and interfaces supporting human-machine real-time collaborative information sharing via multiple media	Integration of human task and machine behavior modeling into common digital data
Digital twin automation	Fully dynamic virtual system copies built from the same models as the real systems running in parallel to physical systems and updating from the same data feeds as their real counterparts	Allows systems to be regularly measured and updated based on learning in new or changing contexts

Table 3.2 Summary of the research in AI/ML relevant to SE disciplines

Research area	Definition	Use in SE
AI/ML Technology Evolution		
Accessibility	AI algorithms and methods become more available in tools that can be used by multiple disciplines	Increase the number of professionals who use and understand the tools
Explainability [DARPA]	Developing sets of machine learning techniques that produce more explainable models, while maintaining a high level of learning performance (prediction accuracy); and enable human users to understand, appropriately trust, and effectively manage the resulting automation	Allowing the human analysis and decisions to better understand and trust the machine-generated analysis and decisions
Cognitive bias	Reducing errors induced in sampled data or algorithms that cause the expected results of the system to be inappropriate for use	Requirements for and evaluation of training data and application usage in the system
Uncertainty quantification (Abdar et al., 2020)	Representing the uncertainty of AI predictions as well as the sources of uncertainty	Requirements for and evaluation of the performance of AI usage in the system
Adversarial attacks (Ren, et al., 2020)	Use of adversarial samples to fool machine learning algorithms; defensive techniques for detection/classification of adversarial samples	Requirements for and evaluation of adversarial defense approaches and their effectiveness in the system
Lifecycle adaptation	Evolution of AI performance over the lifecycle of a system as the system changes/evolves	Learning design and use takes into account the variability of a system over time
AI resilience	Operational resilience of the system and its users incorporating AI, particularly involving the characteristics of ML systems	Application of resilience assessment methods to systems using AI

areas evolving in the current digital transformation of the engineering disciplines (Hagedorn et al., 2020). As more of the underlying data is collected in engineering modeling and analysis, it will become training data for ML applications.

There are several ongoing advancements in digital engineering relevant to AI4SE (McDermott et al., 2020):

Table 3.3 Summary of research in AI/ML relevant to SE disciplines

Research area	Definition	Use in SE
Automation and human–machine teaming		
AI system architecting	Building appropriate data and also live and virtual system architectures to support learning and adaptation and more agile change processes	Parallel development and comparison of in vivo (real) and in silico (virtual) deployments
AI risk analysis	Methods, processes, and tools need to connect system risk analysis results with AI software modules related to those risks	Characteristics of AI systems incorporated into operational loss, hazard, and risk analysis
Calibrating trust	AI systems self-adapt while maintaining rigorous safety, security, and policy constraints	Adaptation and learning incorporated into human system integration
T&E continuum	Methods for addressing AI-related system test and evaluation (T&E) addressing these systems’ ability to adapt and learn from changing deployment contexts	New approaches for both system and user verification and validation (V&V) of adaptive systems
AI/ML at scale	Appreciation for the dependence of an AI’s outputs on its inputs; scale in AI-based systems will increasingly lead to more general intelligence and an inability to relegate AI to a particular subsystem or component	SE frameworks specifying complex system-level behaviors, distinct from decomposition to functions and requirements
Adaptive mission simulation	Computer-based simulation and training supporting non-static objectives and/or goals (games, course-of-action analysis) are necessary to provide contextual learning environments for these systems	Real and simulated co-learning (digital twins) will be a standard system development form

- Tool and Domain Taxonomies and Ontologies: engineering and programmatic data will gain interoperability through domain-specific ontologies. Graph databases for linked data are becoming more prominent in model-based systems engineering tools. Taxonomies provide the starting point for building ontologies, ultimately enabling AI-based reasoning on the underlying data. This advancement is the transformational infrastructure in AI4SE.
- Inter-Enterprise Data Integration: a primary goal of digital engineering is an authoritative source of truth data that underlies the different engineering and

Table 3.4 Summary of research in augmented engineering

Research area	Definition	Use in systems engineering
Augmented engineering		
Automated search	Applying ML to historical data and relationships in the engineering domains	Greatly improve the speed and consistency of systems engineering activities
Conversational data entry	Human/computer interaction processes to convert the natural language and other media to formal models	Improve knowledge transfer and consistency in future systems engineering tools
Automated evidence	Automation of certification and accreditation processes via models, and automation of quality assurance data	Improved speed and coverage, particularly for systems-of-systems and distributed development and test activities
Assurance models	Automation of evidence-based models for assuring correctness and completeness of system requirements and design	Improved specification and verification/validation of critical assurance characteristics
Automated model building/checking	Automated construction of models from features in semantic data used in both creation of new models and correctness of developed models	Improved speed of development
Cognitive assistants	Conversational systems automating many mundane data entry, exploration, and engineering calculation tasks, and many workflows	Improved speed of development, improved collaboration

- program management activities in complex engineered systems. As programs and engineering design activities share data, enterprises will build large datasets for knowledge transfer and reuse across different programs and projects. These data will be available to automate search, model-building/checking, and decision-making.
- **Semantic Rules in Engineering Tools:** based on knowledge representations such as ontologies, semantic rules will provide the basis for reasoning (using AI) about the completeness and consistency of engineering models.
 - **Digital Twin Automation:** engineered systems will be supported by twins—fully dynamic virtual system copies built from the same models as the real systems and running in parallel to physical systems. System design and build data will be updated from the same data feeds as their real counterparts. This dynamic process provides a starting infrastructure for human–machine co-learning.

3.4 AI/ML Technology Evolution

Table 3.2 provides a consolidated summary of the research and development areas in the AI/ML disciplines that are relevant to SE practices. This table is provided as both a view for the SE practitioner as well as perhaps a prioritization in the AI/ML world of research needs, particularly as applications of ML evolve to larger system usage and more critical application areas.

We see a progression of research and development in AI/ML technologies and applications that will lead to increased engineering acceptance and use across more complex engineered systems. This advance starts with the accessibility of AI/ML algorithms and techniques.

The rapid growth of ML technologies has been aided by free open-source tools and low-cost training, but this rise is still targeting computer and data scientists and is based on foundational skills that are not widespread in the SE community. Wade, Buenfil, & Collopy (2020) discuss a potential business model using an abstraction to bring AI/ML to the SE community based on a similar experience in the Very Large Scale Integrated (VLSI) circuit's revolution. Just as abstraction and high-level programming languages hid the underlying complexity of microcircuits from an average designer, the growth in "low-code" AI/ML design tools will make the technology more accessible to other disciplines.

SE is a discipline targeted at improving the predictability of function and performance in the design and use of complex systems. Current day ML applications that "hide" decision paths in deep networks create predictability concerns in the SE community. Even rule-based systems at large scales are a concern for the community, which strives for explicit verification and validation of function and performance in the critical functions of a system. Issues with explainability and data/training bias must be overcome for AI/ML technologies and applications to gain acceptance in the SE community for critical functions. Research in digital twins and extended applications of modeling and simulation for validation are needed. Otherwise, the "validation by use" will be cost and risk-prohibitive in large safety-critical applications. Research in uncertainty quantification of deep learning applications is of particular research interest in the engineering community as certainty in decision-making improves opportunities for validation by decomposition of function. Abdar et al. (2020) provide a good review of this research area.

Improved resilience from design errors and malicious attacks is a concern for use of AI/ML in critical applications. Protection from adversarial attacks and general robustness cannot be provided by add-on applications. It must be designed into the learning process. Ren et al. (McDermott, 2019) provide an overview of this research area and some possible defensive techniques.

In the long term, adaptation and contextual learning in AI/ML systems across long system lifecycles, and the resilience of these systems to changing contexts (environment, use, etc.) will be an active area of research and development in the engineering community. Cody, Adams, and Beling (2020) provide an example of the need and possible approaches to make an AI/ML application more robust to

changes in a physical system over time. This article provides a good example of the challenges of ML in operational environments. Eventually, the broad use of learning applications for multiple interconnected functions in complex systems will arrive. At some point, the SE community will no longer be able to rely primarily on decompositional approaches to system design and must adopt new, more holistic approaches.

Automation and Human–Machine Teaming

Table 3.3 provides a consolidated summary of the research and development areas in automation and human–machine teaming disciplines that are relevant to SE practices. Automation and the use of AI are not new to the SE discipline; but the use of ML is more recent. We envision that humans and machines will team in ways that they learn from each other while using complex engineered systems in complex environments. The robustness of these interactions at scale is an SE challenge.

The future state in SE and automation will see, using a terminology from Madni (2020), the deployment of adaptive cyber-physical-human systems (CPHS). In adaptive CPHS, humans and complex machines learn together as they move across different contexts. Adaptive CPHS employ different types of human and machine learning to flexibly respond to unexpected or novel situations during mission and task execution; to respond using plan and goal adjustment and adaptation; to learn from experience to evolve the system; and to continuously adapt the human and machine tasks in operational performance of the system. A key issue in this future is modeling human behavior in the context of the machine design (Ren et al., 2020). The SE community, to manage complexity and skillsets, generally views the human system activities, the machine function and behavior, and the related modeling and simulation as three independent subdisciplines. These subdisciplines will need to converge as human behaviors and machine behaviors are allowed to adapt together while in use, with the changes sensed by large-scale digital twins. An important research area is adaptive mission simulation—simulation environments that provide contextual learning to both humans and machines across the development, test, and operational lifetimes of a system.

The system architecting process will change as automation scales in more complex systems. System architectures must support learning and adaptation and more agile change processes. System architectures for large hardware systems will include the training data and associated information technologies that support their AI/ML components. Future system architecting must consider the parallel development and the comparison of *in vivo* (real) and *in silico* (virtual) deployments with sensing and data collection subsystems that support continuous learning and adaptation.

The SE community has traditionally viewed T&E of systems and architectures as static events supporting specific lifecycle decisions. Future views of T&E must evolve to support learning and adaptation. Freeman (2020) lists a number of themes for the evolution of the T&E processes to be considered by the SE community:

- T&E is a continuum where data accumulate over time;
- The continuum does not end until the system retires;

- Integrating information from disparate data sources requires unique methods and activities (models, simulations, and test environments) to collect and combine the data;
- Data management is foundational—evaluation of data quality and readiness is essential;
- AI systems require a risk-based test approach that considers all of the evidence collected versus consequence severity in the operational environment;
- Test metrics may have different interpretations, and new metrics may be necessary to focus on risk for AI systems;
- It becomes more essential to understand the operational context and threats for these systems; and to achieve this,
- All AI areas need testbeds for experimentation with operational data.

In all of these cases, pure decomposition of function to design and buildup of function to test may not apply in traditional ways. As multiple AI/ML applications become dependent on each other, the SE community must add methods for the aggregation of decisions and associated behaviors in the systems. Characterization of system behavior in the aggregate will affect traditional T&E approaches as noted previously.

Automation of function has been a continuous feature of engineered systems since the industrial age began, but human–machine co-learning requires different methods to assess risk and trust in future systems. The T&E continuum must support this, but we also need new methods to evaluate risk and to make decisions on whether or not a system is safe, secure, ethical, etc. We are already seeing such issues arise in applications like self-driving cars and facial recognition systems where the societal norms for safety, security, privacy, and fairness are being adjusted. Concepts and metrics for trust need to become more explicit in the SE community—both in the human interaction and the dependability of the machines.

3.5 Augmented Engineering

Table 3.4 provides a consolidated summary of the research and development areas in augmented engineering. We envision that as digital engineering evolves and traditional engineering models and practices rely more on the underlying data, many engineering tasks related to data collection and search, data manipulation, and data analysis will become automated. Also, the machine learning of modeled relationships and underlying data will become more complex over time. This augmentation will automate many mundane engineering tasks leading to a greater focus on problem-solving and design for the human engineer. In addition, we envision that engineering speed and quality will improve as more engineering test and validation activities become automated. The idea of “cognitive assistants” that broadly support the engineer will evolve but they must evolve in a way that supports the problem-solving and associated learning processes associated with engineering.

Rouse (2020) argues for terms like “augmented intelligence” and “augmented engineering” in the SE community because SE is highly associated with human problem-solving. He argues that future cognitive assistants in this domain must not only support automated search and model building/checking but also contextual inferencing of intent, explanation management, and intelligent tutoring with respect to machine inferences and recommendations. Selva and Viros (2019) provide an example of a cognitive assistant for engineering design and system analysis. In this work, they show that a cognitive assistant can increase engineering performance, but, as a side effect, can also decrease human learning (Selva, 2019). This result is an example of the need for co-learning in human–machine teaming.

In the long term, as the engineering community captures more of their process in digital data and models, the use of AI/ML will improve the quality of engineering design and test activities. Automation of data collection and search, model-building, evidence collection, data and model checking, and eventually system assurance processes will lead to better more robust systems.

3.6 Workforce and Culture

In the category of workforce and culture, many system engineers come from foundational disciplines in engineering and lack some of the computer science foundations that drive the AI discipline area. The systems engineering workforce needs to further develop basic digital engineering competencies in software construction and engineering, data engineering, and related information technologies. AI/ML systems are created in these three disciplinary domains. However, SE can bring its strong foundation in interdisciplinary approaches to the AI community. Over time, AI development tools will incorporate design abstractions and patterns that make the technology more accessible to a broad set of engineers, improving the interdisciplinary understanding and use of the technology. A clear workforce development concern is the integration of AI with systems engineering and human systems integration—a much greater representation of the cognitive sciences and cognitive engineering in the SE discipline set. Specialty systems engineering disciplines such as security and safety must move to the forefront. New test and evaluation approaches for learning and adaptation will significantly affect those disciplines.

3.7 Summary—The AI imperative for Systems Engineering

SE is undergoing a digital transformation. This evolution will lead to further transformational advances in the use of AI and ML technology to automate many routine engineering tasks. At the same time, applying AI, ML, and autonomy to complex and critical systems encourages new systems engineering methods, processes, and tools. It is imperative that the systems engineering community deeply understand

the emerging AI and ML technologies and applications, incorporate them into the methods and tools in ways that improve the SE discipline, and ensure that appropriate systems engineering approaches are used to make AI systems ethical, reliable, safe, and secure. The road mapping activity presented here attempts to understand broadly all of the opportunities and risks that might appear as this evolution proceeds as well as potentially provide the information that guides further research.

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